# Second-Order ODEs

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# Second Order ODEs

- Any second order differential equation can be written as

$$F(x, y, y', y'') = 0.$$

$e^{y''} + \sin xy = e^x$

- We are mainly concerned with very important second-order equations, namely linear equations. Recall that a first-order linear differential equation is an equation which can be written in the form $y' + p(x)y = q(x)$, where $p$ and $q$ are continuous functions on some interval $I$. A second-order, linear differential equation has an analogous form.

## Definition (Second-Order Linear ODEs)

A second-order linear differential equation is an equation that can be written in the form

$$\underbrace{y'' + p(x)y' + q(x)y}_{\text{linear in } y, y', y''} = f(x)$$

$$\Rightarrow \overbrace{y'' = -p(x)y' - q(x)y + f(x)}^{\text{Normal form}} = F(x, y, y')$$

where $p, q$, and $f$ are continuous functions on some interval $I$.

# Second-Order ODEs

$y'' = f(x, y, y')$

• The functions $p$ and $q$ are called the coefficients of the equation; the function $f$ on the right-hand side is called the forcing function or the nonhomogeneous term. $(f(x) \neq 0)$

A second-order equation which is not linear is said to be **nonlinear**.

•The equation

$$x^2y'' - 2xy' + 2y = 0 \Rightarrow y'' - \frac{2}{x}y' + \frac{2}{x^2}y = 0$$

$(x \neq 0)$

$p(x) = -\frac{2}{x}$, $q(x) = \frac{2}{x^2}$, $f(x) = 0$ Con t $x \neq 0$

is a second-order linear ODE.

• The equation

$$y'' + \underbrace{xy^2y' - y^3}_{\text{linearity fails.}} = e^x y$$

is a nonlinear second-order ODE.

# Reason for Calling it *Linear*

$L(y)=0. \Longleftrightarrow y''+p(x)y'+q(x)y=0$

- Set $L[y] = y'' + p(x)y' + q(x)y$. If we view $L$ as an "operator" that transforms a twice differentiable function $y = y(x)$ into the continuous function

$$\boxed{L[y(x)] = y''(x) + p(x)y'(x) + q(x)y(x)}.$$

- Then, for any two twice differentiable functions $y_1(x)$ and $y_2(x)$,

$$L[y_1(x) + y_2(x)] = [y_1(x) + y_2(x)]'' + p(x)[y_1(x) + y_2(x)]' \\ + q(x)[y_1(x) + y_2(x)]$$

$$= \left(y_1''(x) + p(x)y_1'(x) + q(x)y_1(x)\right) + \left(y_2''(x) + p(x)y_2'(x) + q(x)y_2(x)\right)$$

$$= L(y_1(x)) + L(y_2(x))$$

# Reason for Calling it *Linear*. Contd.

- Also, for any constant $c$,

$$L[cy_1(x)] = [cy_1(x)]'' + p(x)[cy_1(x)]' + q(x)[cy_1(x)]$$

$$= c\left(y_1''(x) + p(x)y_1'(x) + q(x)y_1(x)\right)$$

$$= cL(y_1)$$

$y_1, y_2 \in C$, $c \in \mathbb{R}$
$$\left.\begin{cases} L(y_1+y_2) = L(y_1) + L(y_2) \\ L(cy_1) = cL(y_1) \end{cases}\right\} \text{linear}$$

Rem :- Sol$^n$ of $y'' + p(x)y' + q(x)y = 0$ will form a vector space.

# Existence & Uniqueness Theorem

$y'' = F(x, y, y') \longrightarrow$ Explore

**Theorem (Existence and Uniqueness Theorem)**

*Given the second-order linear equation*

$$y'' + p(x)y' + q(x)y = f(x) \text{ in } I,$$

*where $p, q, f$ are continuous functions over the interval $I$. Let $a$ be any point on the interval $I$, and let $\alpha$ and $\beta$ be any two real numbers. Then the initial-value problem*

$$y'' + p(x)y' + q(x)y = f(x), \; y(a) = \alpha, \; y'(a) = \beta$$

*has a unique solution.*

$y' + p(x)y = q(x)$

$y'' = f(x) - p(x)y' - q(x)y = F(x, y, y')$

$\boxed{y'' = F(x, y, y')}$

$\boxed{\frac{\partial F}{\partial y} = -q(x) \quad \Big| \quad \frac{\partial F}{\partial y'} = -p(x)}$

## Example

Consider the IVP:

$$(x-1)y'' + 3xy' + 4y = \sin x, \;\; y(0) = 1, \;\; y'(0) = 0.$$

$$\Rightarrow \quad y'' + \frac{3x}{x-1}\, y' + \frac{4}{x-1}\, y = \frac{\sin x}{x-1}, \quad (x \neq 1)$$

$$\left.\begin{aligned} p(x) &= \frac{3x}{x-1} \\ q(x) &= \frac{4}{x-1} \\ f(x) &= \frac{\sin x}{x-1} \end{aligned}\right\}$$

Let $I = (-\infty, 1) \ni 0$

$p, q, f$ are not $\mathrm{cont}^{s}$ at $\boxed{x=1}$

Existence & Uniqueness is ensured.

# Second Order Linear Homogeneous Equations

$y \neq 0 \dashv$ ~~$y(x) \neq 0$~~, $\forall x \in \mathbb{R}$ $\quad$ $\boxed{y \equiv 0, \quad y(x) = 0, \text{ for all } x \in I}$

• As defined in the previous section, a second-order linear homogeneous differential equation is an equation that can be written in the form

$$y'' + p(x)y' + q(x)y = 0$$

where $p$ and $q$ are continuous functions on some interval $I$.

• **The trivial solution.** The first thing to note is that the zero function, $y(x) = 0$ for all $x \in I$, (also denoted by $y \equiv 0$) is a solution of the above homogeneous ODE ($y \equiv 0 \implies y' \equiv 0$ and $y'' \equiv 0$).

• The zero solution is called the trivial solution. Obviously, our main interest is in finding nontrivial solutions. Unless specified otherwise, the term "solution" will mean "nontrivial solution."

# Vector Space Structure of Solutions of Linear Homogeneous ODEs

## Theorem (Superposition Principle)

*If $y = y_1(x)$ and $y = y_2(x)$ are any two solutions of*

$$y'' + p(x)y' + q(x)y = 0,$$

*and if $C_1$ and $C_2$ are any two real numbers, then*

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

*is also a solution.*

vector space structure

## Remark

Any linear combination of solutions of a linear homogeneous equation is also a solution.

**Question**

Suppose that $y = y_1(x)$ and $y = y_2(x)$ are solutions of the homogeneous equation $y'' + p(x)y' + q(x)y = 0$. Under what conditions is $y(x) = C_1 y_1(x) + C_2 y_2(x)$ the general solution of the linear homogeneous ODE?

$L(y) = y'' + p(x) y' + q(x) y$ Null$(L)$

$\dim(\text{Null}(L)) = ?$ (Ans 2)

# Origin of the Idea of the Wronskian

• Let $u = u(x)$ be any solution of $y'' + p(x)y' + q(x)y = 0$ and choose any point $a \in I$. Suppose that $\alpha = u(a)$, $\beta = u'(a)$.

• Then $u$ is a member of the two-parameter family $y(x) = C_1 y_1(x) + C_2 y_2(x)$ if and only if there are values for $C_1$ and $C_2$ such that

$$C_1 y_1(a) + C_2 y_2(a) = \alpha \qquad \longrightarrow \times y_1'(a)$$

and

$$C_1 y_1'(a) + C_2 y_2'(a) = \beta. \qquad \longrightarrow \times y_1(a)$$

Since $y_1$ and $y_2$ are sol$^n$s we have

$$\left.\begin{aligned} &y_1'' + p(x) y_1' + q(x) y_1 = 0, \quad y_1(a) = \alpha \text{ and } y_1'(a) = \beta \\ &y_2'' + p(x) y_2' + q(x) y_2 = 0, \quad y_2(a) = \alpha \text{ and } y_2'(a) = \beta \end{aligned}\right\}$$

So $Ly = y'' + p(x)y' + q(x)y$ gives us

$L(c_1y_1 + c_2y_2) = c_1L(y_1) + c_2L(y_2) = (c_1 \times 0) + (c_2 \times 0) = 0.$

Therefore $c_1y_1 + c_2y_2$ be a sol$^n$ one criteria

$L(c_1y_1 + c_2y_2) = 0$

is fullfilled. Only auxiliary cond$^n$s need to be matched.

So we must have: $(c_1y_1 + c_2y_2)(a) = \alpha$ and $(c_1y_1 + c_2y_2)'(a) = \beta$

Qn: Can we solve for unknowns $c_1$ and $c_2$?

Multiplying the first by $y_1'(a)$ and second by $y_1(a)$ and subtracting, we get

$$\left.\begin{aligned} c_1y_1(a)y_1'(a) + c_2y_2(a)y_1'(a) &= \alpha y_1'(a) \\ \underline{-}\ c_1y_1'(a)y_1(a) \ \underline{\pm}\ c_2y_2'(a)y_1(a) &= \underline{-}\ \beta y_1(a) \end{aligned}\right\}$$

$$\Rightarrow c_2\, y_2(a)\, y_1'(a) - c_2\, y_2'(a)\, y_1(a) = \alpha\, y_1'(a) - \beta\, y_1(a)$$

$$\Rightarrow \boxed{c_2 = \frac{\alpha\, y_1'(a) - \beta\, y_1(a)}{y_2(a)\, y_1'(a) - y_2'(a)\, y_1(a)}}$$

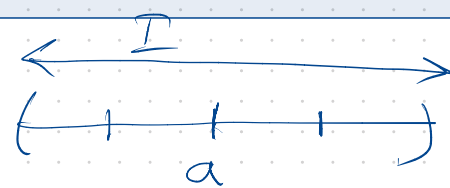
Similarly, we get

$$\boxed{c_1 = \frac{-\alpha\, y_2'(a) + \beta\, y_2(a)}{y_2(a)\, y_1'(a) - y_2'(a)\, y_1(a)}}$$

Now, $c_1$ and $c_2$ make sense provided

$$\boxed{y_2(a)\, y_1'(a) - y_2'(a)\, y_1(a) \neq 0}$$

$$\begin{array}{l|l} y'' + p(x)\,y' + q(x)\,y = 0 & \mathrm{I} \\ y(a) = \alpha, \quad y'(a) = \beta & \mathrm{V} \\ a \in I & \mathrm{P} \end{array}$$



# Wronskian

- Since $a$ was chosen to be any point on $I$, we conclude that $y(x)$ is the general solution of the homogeneous equation if

$$y_1(x)y_2'(x) - y_2(x)y_1'(x) \neq 0 \ \forall \ x \in I.$$

$$\begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} \neq 0$$

## Theorem (Wronskian)

*Let $y = y_1(x)$ and $y = y_2(x)$ be solutions of $y'' + p(x)y' + q(x)y = 0$. The function $W$ is defined by $W[y_1, y_2](x) = y_1(x)y_2'(x) - y_2(x)y_1'(x)$ is called the Wronskian of $y_1, y_2$.*

## Remark

We use the notation $W[y_1, y_2](x)$ to emphasize that the Wronskian is a function of $x$ that is determined by two solutions $y_1, y_2$ of linear homogeneous equation. When there is no danger of confusion, we'll shorten the notation to $W(x)$.

# General Solution & Wronskian

## Theorem (Wronskian)

*Let* $y = y_1(x)$ *and* $y = y_2(x)$ *be solutions of equation*

$$y'' + p(x)y' + q(x)y = 0,$$

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}$$

*and let* $W(x)$ *be their Wronskian. Exactly one of the following holds:*

(i) $W(x) = 0$ *for all* $x \in I$; $y_1$ *is a constant multiple of* $y_2$ *and vice versa.*

(ii) $W(x) \neq 0$ *for all* $x \in I$ *and* $y = C_1y_1(x) + C_2y_2(x)$ *is the general solution of the above linear homogeneous ODE.*

# Example: General Solution

## Example (General Solution)

Verify $y_1(x) = e^x$ and $y_2(x) = e^{2x}$ are each solutions of

$$y'' - 3y' + 2y = 0.$$

We want to determine whether or not the two-parameter family

$$y(x) = C_1 e^x + C_2 e^{2x} \quad = c_1 y_1(x) + c_2 y_2(x)$$

is the general solution of the above homogeneous linear second-order ODE.

$$\left.\begin{aligned} (e^x)'' - 3(e^x)' + 2e^x &= 0 \\ (e^{2x})'' - 3(e^{2x})' + 2e^{2x} &= 0 \end{aligned}\right\} \checkmark$$

$$W[y_1, y_2](x) = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} \neq 0. \quad \forall x \in I$$

# Example: General Solution

## Example

Verify the functions $y_1(x) = x^2$ and $y_2(x) = 5x^2$ are solutions of

$$x^2y'' - 2xy' + 2y = 0.$$

Also, check that whether or not the two-parameter family

$$y = C_1x^2 + C_2(5x^2)$$

is the general solution.

$x^2$ and $5x^2$ will solve the ODE.

Qn. Is $c_1x^2 + c_2(5x^2)$ a general sol$^n$?

$W = 0 \Rightarrow$ not a general sol$^n$ (full-calculation)

Qn:- What is the general sol$^n$'s form?

⊙ There is a connection between Wronskian and linear dependence/independence.

**Theorem:** Let $f = f(x)$ and $g = g(x)$ be diff$^{ble}$ f$^n$s on an open interval $I$. If $f$ and $g$ are linearly dependent on $I$, then $W(x) = 0$ for all $x \in I$. ($W = 0$ on $I$).

<u>Remark</u>:- Equivalently, let $f = f(x)$ and $g = g(x)$ be diff$^{ble}$ f$^n$ on an interval $I$. If $W(x) \neq 0$ for at least one $x \in I$, then $f$ and $g$ are linearly independent.

Going back to differential equation, the theorem on Wronskian can be restated as follows:

**Theorem:** Let $y = y_1(x)$ and $y = y_2(x)$ be sol$^{\underline{n}}$ of the homogeneous equation

$$y'' + p(x)\,y' + q(x)\,y = 0.$$

Then exactly one of the following holds:

(i) $W(x) = 0$, $\forall\, x \in I$, $y_1$ and $y_2$ are dependent.

(ii) $W(x) \neq 0$, $\forall\, x \in I$; $y_1$ and $y_2$ are linearly independent and $y = c_1 y_1(x) + c_2 y_2(x)$ is the general sol$^{\underline{n}}$.

# Thank You

Questions?